

Problem Set 3.1 – Sensor Anomaly Filter

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Source Code

```
1  # Logic Trace: a for-loop is bounded by the length of what it iterates
2  # over. If a batch is [], there's nothing to loop through, so the body
3  # just never runs and we move to the next batch. No way for it to hang.
4  #
5  # Author: Kshitiz
6
7  telemetry_stream = [
8      [22.5, 23.0, 22.8],
9      [25.1, "ERR", 24.9],
10     [30.2, 35.5, 40.1],
11     [22.0, 22.1, "STOP"],
12 ]
13
14 shutdown_triggered = False
15
16 for batch_id in range(len(telemetry_stream)):
17     batch = telemetry_stream[batch_id]
18     previous_value = None
19     print(f"--- Auditing Batch {batch_id}: {batch} ---")
20
21     for reading in batch:
22         if reading == "STOP":
23             print(f"Emergency Shutdown at Batch {batch_id}.")
24             shutdown_triggered = True
25             break
26
27         if reading == "ERR":
28             print(f"Noise ignored at Batch {batch_id} (ERR).")
29             continue
30
31         if isinstance(reading, (int, float)):
32             if reading > 35.0:
33                 print(f"Anomaly Detected at Batch {batch_id}: {reading}")
34
35             if previous_value is not None:
36                 delta = abs(reading - previous_value)
37                 if delta > 5.0:
38                     print(f"Spike Detected at Batch {batch_id}: "
39                           f"{previous_value} -> {reading} (Delta {delta:.1f})")
40
41             previous_value = reading
42
43     if shutdown_triggered:
44         break
45
46 else:
47     print("Audit Complete: No system-wide failures")
48
49 # Submission Note: shutdown_triggered starts False and only gets set to
50 # True when "STOP" shows up. break only exits the inner loop, so right
```

```
51 # after it I check the flag and break the outer loop too - that's what
52 # actually stops the audit. The outer else only runs if the outer loop
53 # wasn't broken, so it correctly skips "Audit Complete" whenever STOP
54 # was seen, and prints it normally otherwise.
```

Trace Log

The script was run against the sample telemetry stream from the problem statement. Output below matches the reference trace given in Section 6 of the assignment.

```
1 --- Auditing Batch 0: [22.5, 23.0, 22.8] ---
2 --- Auditing Batch 1: [25.1, 'ERR', 24.9] ---
3 Noise ignored at Batch 1 (ERR).
4 --- Auditing Batch 2: [30.2, 35.5, 40.1] ---
5 Anomaly Detected at Batch 2: 35.5
6 Spike Detected at Batch 2: 30.2 -> 35.5 (Delta 5.3)
7 Anomaly Detected at Batch 2: 40.1
8 --- Auditing Batch 3: [22.0, 22.1, 'STOP'] ---
9 Emergency Shutdown at Batch 3.
```

No `Audit Complete` line appears because `STOP` was hit in Batch 3, so the outer loop's `else` clause is correctly skipped.

Edge Cases Verified

- **Empty batch** – an empty inner list produces no output for that batch and the audit continues normally; the loop simply has zero elements to step through.
- **All-control batch** (`["ERR", "ERR"]`) – both readings are skipped via `continue`, `previous_value` stays `None`, and no delta comparison is attempted.
- **STOP as first reading** – shutdown logic fires immediately, even though no numeric reading was processed in that batch yet.

Submission Notes

The trickiest part of this problem was making sure `break` inside the inner loop didn't just quietly let the outer loop move on to the next batch. Since `break` only kills the loop it's written in, I used a `shutdown_triggered` flag that gets set the moment `STOP` shows up, then checked right after the inner loop ends so the outer loop can break too. That flag is also what the outer `else` depends on – it only prints `Audit Complete` when the outer loop runs to completion without being broken, which is exactly the case where `STOP` was never seen.

For the bonus Rolling Delta check, `previous_value` is reset to `None` at the top of every batch (inside the outer loop, outside the inner one) so a spike from one batch never bleeds into the next. It only updates after a genuine numeric reading, so an `ERR` in between two real values doesn't reset the delta tracking by accident.